🔑 **Answer Key — Teacher Copy · Lesson 6-13**

MSTAR PRACTICE WORKSHEET

Lesson 6-13: Prime Factorization

Grade 6 Mathematics · Standard 6.NOS.4

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: C

Which of the following is a prime number?

A. 9

B. 15

C. 17

D. 21

17 has exactly two factors: 1 and 17. 15 = 3 × 5, 21 = 3 × 7, and 9 = 3 × 3, so they are all composite.

Question 2 SELECTED RESPONSE — CORRECT: A

What is the prime factorization of 30?

A. 2 × 3 × 5

B. 2 × 15

C. 3 × 10

D. 5 × 6

30 = 2 × 15 = 2 × 3 × 5. All three factors (2, 3, 5) are prime, so 2 × 3 × 5 is the prime factorization.

Question 3 SELECTED RESPONSE — CORRECT: A

What is the prime factorization of 18?

- (A) 2 × 3 × 3
- (B) 2 × 9
- (C) 3 × 6
- (D) 6 × 3

18 = 2 × 9 = 2 × 3 × 3. Both 2 and 3 are prime, so 2 × 3 × 3 is the prime factorization.

Question 4 SELECTED RESPONSE — CORRECT: B

Which of these numbers is composite?

- (A) 23
- (B) 27
- (C) 29
- (D) 31

27 = 3 × 9 = 3 × 3 × 3, so it has more than two factors. 23, 29, and 31 are all prime.

Question 5 SELECTED RESPONSE — CORRECT: D

Two students found different factor trees for 60. Student A started with 2 × 30. Student B started with 6 × 10. Which statement is true?

- (A) Only Student A gets the correct prime factorization
- (B) Only Student B gets the correct prime factorization
- (C) They will get different prime factorizations
- (D) Both get the same prime factorization: 2 × 2 × 3 × 5

The Fundamental Theorem of Arithmetic says every composite number has exactly one prime factorization. No matter how you start the factor tree, you always end with 2 × 2 × 3 × 5.

Question 6 SELECTED RESPONSE — CORRECT: A

The first lock panel shows 90. What is the prime factorization of 90?

- (A) 2 × 3 × 3 × 5
- (B) 2 × 45
- (C) 3 × 30
- (D) 9 × 10

90 = 2 × 45 = 2 × 9 × 5 = 2 × 3 × 3 × 5. Every factor is prime, so the lock accepts 2 × 3 × 3 × 5.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: A

What is the prime factorization of 72?

(A) $2^3 \times 3^2$

(B) $2^2 \times 3^3$

(C) $2 \times 3 \times 12$

(D) 8×9

A factor tree for 72 ends in the primes 2, 2, 2, 3, and 3. Written with exponents in ascending order, that is $2^3 \times 3^2$, and $8 \times 9 = 72$ confirms the value.

- **B:** The exponents are swapped: four times twenty-seven is 108. Seventy-two holds three 2s and two 3s.
- **C:** 12 is not prime. Keep factoring it into 2 times 2 times 3.
- **D:** Both factors are right but neither is prime. Break 8 into three 2s and 9 into two 3s.

Part B — correct answer: B

Which reasoning shows why your answer to Part A is correct?

(A) The tree gives two 2s and three 3s, so the prime factorization is $2^2 \times 3^3$.

(B) Every branch of the factor tree for 72 ends in a prime: 2, 2, 2, 3, 3. Grouping the repeated primes with exponents gives $2^3 \times 3^2$, and $2^3 \times 3^2 = 8 \times 9 = 72$.

(C) 72 splits into $2 \times 3 \times 12$, and the branches can stop there.

(D) 72 splits into 8×9, and those are the factors to report.

A prime factorization must have every branch ending in a prime number. The number 12 is composite, and 8 and 9 are both composite, so those splits are unfinished. Counting the primes gives three 2s and two 3s.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, E

Select ALL statements that are true about the number 45.

- **A** 45 is a composite number.
- **B** The prime factorization of 45 is $3^2 \times 5$.
- **C** The prime factorization of 45 is 5×9.
- **D** 45 is a prime number.
- **E** The only prime factors of 45 are 3 and 5.
- **F** 1 is a prime factor of 45.

45 has the factors 1, 3, 5, 9, 15, and 45, so it is composite, not prime. Its factor tree ends in the primes 3, 3, and 5, giving $3^2 \times 5 = 9 \times 5 = 45$. The expression 5×9 is not a prime factorization because 9 is composite, and 1 is never a prime factor because a prime has exactly two factors.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

Choose any two-digit composite number. Show TWO different factor trees that both lead to the same prime factorization. Explain why every composite number has only one prime factorization.

Model answer: A strong answer picks one two-digit composite number and finishes two trees that begin with different first splits — for 24, starting with 2×12 and starting with 4×6 both end at $2 \times 2 \times 2 \times 3$. The lists always match because splitting only pulls composite factors apart into primes that were already inside the number, so the set of primes cannot change.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.